

Notes: He, Su & Yu (JFE, 2024)
Macroeconomic Perceptions, Financial Constraints, and Anomalies

Contents

1	Big picture	3
2	Data and measurement	3
2.1	Anomalies and financial-constraint classification	3
2.2	Financial-constraint indices	4
2.3	Macroeconomic expectation revisions	4
2.4	Anomaly returns and controls	4
2.5	Mechanism variables	5
3	Models and empirical specifications	5
3.1	Two-regime portfolio-return test	5
3.2	Baseline predictive regression	5
3.3	Horse-race predictive regression	5
3.4	Misperception regression	6
3.5	Corporate-activity regression	6
3.6	Aggregate predictability regression	6
4	Identification and empirical design	6
4.1	What identifies the mechanism	6
4.2	Why forecast revisions are useful	7
4.3	Alternative explanations addressed	7
4.4	What the paper cannot fully prove	7
5	Tables and figures	8
5.1	Table 1 and Figure 1: anomaly classification and IPG forecast revisions	8
5.2	Table 2 and Table 3: anomaly summaries and key variables	9
5.3	Table 4 and Table 5: two-regime anomaly returns	9
5.4	Figure 2 and Table 6: alternative surveys and predictive regressions	10
5.5	Table 7: controlling for confounding variables	12
5.6	Figure 3 and Table 8: rotation strategy and misperception evidence	13
5.7	Table 9 and Table 10: corporate activities and constraint-sorted portfolios	14
5.8	Table 11: aggregate return and credit predictability	15
6	Short summary	17
7	Conclusion	17
7.1	Core conclusion	17

7.2	Economic intuition	17
7.3	Takeaway	17

1 Big picture

- **Question.** How do subjective macroeconomic expectations affect the cross-section of anomaly returns, and why are the effects stronger for financially constrained firms?
- **Main explanatory variable.** The paper uses revisions in Survey of Professional Forecasters expectations of one-quarter-ahead industrial production growth, denoted Frev_IPG_t , as a proxy for changes in investors' perceptions of aggregate productivity.
- **Core idea.** Upward revisions in expected productivity can make managers and investors overly optimistic. This leads to more external financing, more investment, lower financing costs, inflated current prices, and lower subsequent returns for firms that are more financially constrained.
- **Anomaly split.** The paper studies 12 anomalies. In the profitability, quality, distress, and low-risk categories, the *short legs* are more financially constrained. In the value, investment, and size categories, the *long legs* are more financially constrained.
- **Prediction.** After upward IPG forecast revisions, anomalies whose short legs are more constrained become stronger, because the constrained short-leg firms underperform. Anomalies whose long legs are more constrained become weaker or reverse, because the constrained long-leg firms underperform.
- **Opposite regime.** After downward IPG forecast revisions, constrained firms perform better relative to unconstrained firms. Thus value, investment, and size anomalies strengthen when the macro expectation revision is downward.
- **Main return evidence.** In the two-regime tests, the average long-short return for profitability, quality, distress, and low-risk anomalies is 3.73% per quarter after upward IPG revisions and -0.35% after downward revisions. For value, investment, and size anomalies, the corresponding averages are -1.30% and 2.34%.
- **Predictive-regression evidence.** One-standard-deviation increases in IPG forecast revisions predict stronger profitability/quality/distress/low-risk anomalies and weaker value/investment/size anomalies.
- **Mechanism evidence.** Upward IPG revisions predict systematic errors in analysts' long-term earnings growth expectations, especially for financially constrained firms.
- **Corporate-activity evidence.** Following upward IPG revisions, constrained firms increase external financing and investment more, and their financing conditions appear to ease more.
- **Factor-investing implication.** A strategy that rotates between anomaly groups based on the previous quarter's IPG forecast revision outperforms the static anomaly groups in cumulative return.
- **Economic takeaway.** Macroeconomic belief revisions interact with financial constraints. This interaction generates predictable time variation in anomaly returns and provides a behavioral mechanism for why constrained firms' prices can overreact to macroeconomic news.

2 Data and measurement

2.1 Anomalies and financial-constraint classification

The paper studies 12 anomalies. Table 1 divides them into two groups according to which side of the long-short portfolio is more financially constrained.

- **Short leg more constrained:** return on assets, return on equity, operating profitability, quality, failure probability, O-score, total volatility, and idiosyncratic volatility.
- **Long leg more constrained:** book-to-market equity, long-term reversals, asset growth, and market capitalization.
- The financial-constraint ordering follows prior literature and is verified with constructed composite financial-constraint indices.

Interpretation: The central empirical design depends on knowing which leg of each anomaly is more financially constrained. Once this is known, the predicted sign of the response to macro expectation revisions is clear.

2.2 Financial-constraint indices

The paper constructs a composite financial-constraint index from three established measures: the Whited-Wu index, the Hadlock-Pierce index, and the Kaplan-Zingales index.

The Whited-Wu index is

$$WW_{it} = -0.091CF_{it} - 0.062DIVPOS_{it} + 0.021LTD_{it} - 0.044LNTA_{it} + 0.102ISG_{it} - 0.035SG_{it}. \quad (1)$$

The Hadlock-Pierce index is

$$HP_{it} = -0.737LNTA_{it} + 0.043LNTA_{it}^2 - 0.040AGE_{it}. \quad (2)$$

The Kaplan-Zingales index is

$$KZ_{it} = -1.001909CF_{it} + 0.2826389Q_{it} + 3.139193LEV_{it} - 39.3678DIV_{it} - 1.314759CASH_{it}. \quad (3)$$

Interpretation: These indices capture financing frictions from different angles: internal cash flow, dividend status, leverage, size, age, investment opportunities, and cash holdings. The paper averages cross-sectional ranks to form a quarterly composite index.

2.3 Macroeconomic expectation revisions

The key macro perception variable is the revision in expectations of one-quarter-ahead industrial production growth:

$$\text{Frev_IPG}_t = E_t[IPG_{t+1}] - E_{t-1}[IPG_{t+1}]. \quad (4)$$

- The expectations come from the Survey of Professional Forecasters.
- A positive revision means professional forecasters expect higher near-term industrial production growth than they did in the prior quarter.
- The paper standardizes forecast revisions to have mean zero and unit variance in regressions.
- The sample for most tests spans 1969Q1 to 2019Q4, with some anomaly samples starting later.

Interpretation: The variable is forward-looking. It captures changes in perceived aggregate productivity rather than realized macroeconomic conditions.

2.4 Anomaly returns and controls

- For each anomaly, the paper reports long-leg returns, short-leg returns, and long-short spreads.
- The baseline returns are quarterly excess returns.
- Robustness checks use CAPM alphas and Fama-French five-factor alphas.
- Predictive regressions often include the contemporaneous market factor $Mktrf_{t+1}$ to isolate anomaly-specific predictive power.
- Further regressions control for sentiment, surplus consumption ratio, degree of extrapolative weighting, return dispersion, value spread, realized volatility, and realized macro shocks.

Interpretation: The point is not only that IPG forecast revisions predict returns, but that they do so in the sign pattern implied by the financial-constraint channel.

2.5 Mechanism variables

The paper examines three mechanism groups.

- **Misperception:** errors in analysts' long-term earnings growth forecasts.
- **Corporate activities:** changes in bond yields, debt, long-term debt, equity issuance, investment, and financial-constraint indices.
- **Aggregate predictability:** stock market returns, bond market returns, and future bank credit tightening.

Interpretation: The mechanism evidence asks whether the return patterns are linked to expectation errors and real financing/investment responses rather than only to statistical return predictability.

3 Models and empirical specifications

3.1 Two-regime portfolio-return test

The first empirical design splits quarters into upward and downward IPG forecast revisions and estimates

$$R_{t+1} = a_U \mathbf{1}_{U,t} + a_D \mathbf{1}_{D,t} + \varepsilon_{t+1}, \quad (5)$$

where $\mathbf{1}_{U,t}$ and $\mathbf{1}_{D,t}$ are indicators for upward and downward forecast revisions.

Interpretation: The coefficient a_U is the average next-quarter return after an upward productivity-expectation revision. The coefficient a_D is the average next-quarter return after a downward revision. The difference $a_U - a_D$ is the regime effect.

3.2 Baseline predictive regression

The paper then estimates

$$R_{t+1} = \alpha + \beta \text{Frev_IPG}_t + \gamma \text{Mktrf}_{t+1} + \varepsilon_{t+1}. \quad (6)$$

- R_{t+1} is the excess return on the anomaly long leg, short leg, or long-short spread.
- Frev_IPG_t is standardized.
- Mktrf_{t+1} controls for the market factor.

Interpretation: The sign of β should depend on which side of the anomaly portfolio is more financially constrained. A positive β is expected for the long-short spread when the short leg is constrained; a negative β is expected when the long leg is constrained.

3.3 Horse-race predictive regression

The paper controls for candidate alternative predictors using

$$R_{t+1} = \alpha + \beta \text{Frev_IPG}_t + \gamma z_t + \varepsilon_{t+1}, \quad (7)$$

where z_t is a control variable such as investor sentiment, surplus consumption ratio, return dispersion, value spread, realized volatility, or realized macro shocks.

Interpretation: The objective is to test whether subjective productivity-expectation revisions contain distinct information not captured by familiar business-cycle, sentiment, or risk variables.

3.4 Misperception regression

For analysts' long-term earnings growth forecasts, the forecast error is

$$LTG_Error_{t+1} = \left(\frac{EPS_{t+13}}{EPS_{t+1}} \right)^{1/3} - LTG_{t+1}. \quad (8)$$

The test regression is

$$LTG_Error_{t+1} = \alpha + \beta Frev_IPG_t + \varepsilon_{t+1}. \quad (9)$$

Interpretation: A negative coefficient means that upward IPG forecast revisions lead analysts to overpredict future long-term earnings growth. The paper finds that this pattern is especially pronounced for the financially constrained legs of the anomaly portfolios.

3.5 Corporate-activity regression

To study real and financing responses, the paper estimates

$$Y_t = \alpha + \beta Frev_IPG_t + \varepsilon_t, \quad (10)$$

where Y_t includes changes in bond yield, debt, long-term debt, equity, investment, and the composite financial-constraint index.

Interpretation: If investors and managers overreact to good macro news, financially constrained firms should experience larger easing in financing conditions and larger increases in external financing and investment.

3.6 Aggregate predictability regression

The paper also regresses aggregate stock returns, corporate bond returns, and bank credit conditions on forecast revisions:

$$Y_{t+h} = \alpha + \beta Frev_IPG_t + \varepsilon_{t+h}. \quad (11)$$

Interpretation: This tests whether the overreaction mechanism appears beyond cross-sectional anomalies. The paper finds that IPG forecast revisions predict bond-market returns and credit conditions more clearly than aggregate stock market returns.

4 Identification and empirical design

4.1 What identifies the mechanism

- The paper does not rely on a single treatment event. It uses time-series variation in subjective macroeconomic expectation revisions.
- The key identifying pattern is *sign-reversal across anomaly groups*: the same IPG revision should strengthen one set of anomalies and weaken the other set because the constrained leg switches sides.
- The evidence is strengthened by showing similar effects in returns, predictive regressions, analyst forecast errors, firm financing and investment, and aggregate credit conditions.

Interpretation: A generic macro risk story would struggle to explain why IPG forecast revisions predict opposite signs depending on whether the constrained firms are in the long or the short leg.

4.2 Why forecast revisions are useful

- Forecast revisions are forward-looking and capture changes in perceived macroeconomic productivity.
- They avoid using realized macro variables that may be noisy or only known after the return period.
- The SPF release timing helps align revisions with the quarter in which expectations change.

Interpretation: The variable measures perceptions, not fundamentals alone. This makes it suitable for testing overreaction and misperception mechanisms.

4.3 Alternative explanations addressed

The paper conducts several robustness and horse-race exercises.

- It controls for the market factor and uses CAPM-adjusted returns.
- It compares IPG forecast revisions with realized productivity shocks.
- It controls for investor sentiment, surplus consumption ratio, degree of extrapolative weighting, return dispersion, value spread, and realized volatility.
- It controls for a broad set of macro predictors in appendix tests.
- It uses alternative macro expectation measures, including unemployment-rate expectations and survey measures from consumers and CFOs.

Interpretation: The central result remains that subjective productivity forecast revisions predict anomaly returns with the sign pattern implied by financial constraints.

4.4 What the paper cannot fully prove

- Survey forecast revisions are proxies for investor and manager perceptions; the paper cannot observe all market participants' beliefs directly.
- Financial-constraint indices are noisy proxies for true financing frictions.
- The paper provides evidence consistent with overreaction, but it cannot directly observe every belief-updating channel behind trading decisions.
- Some results rely on portfolio-level aggregation, which may hide firm-level heterogeneity.

Interpretation: The evidence is strongest as a coherent pattern across multiple outcomes rather than as a single causal experiment.

5 Tables and figures

This section keeps the compact visual layout, but adds more interpretation for each table and figure. The main reading principle is: always ask which portfolio leg is more financially constrained, then translate the sign of the IPG forecast revision into expected overreaction or reversal for that leg.

5.1 Table 1 and Figure 1: anomaly classification and IPG forecast revisions

Read:

- Table 1 is not just descriptive; it sets up the paper’s sign predictions. The anomalies are sorted by whether constrained firms appear in the long leg or the short leg.
- The short-leg constrained group includes profitability, quality, distress, and low-risk anomalies. If constrained short-leg firms underperform after macro optimism, the long-short anomaly return should rise.
- The long-leg constrained group includes value, long-term reversals, investment, and size. If constrained long-leg firms underperform after macro optimism, the long-short anomaly return should fall or become negative.
- Figure 1 plots the SPF revision in one-quarter-ahead industrial production growth expectations. Positive values are upward revisions in perceived future productivity; negative values are downward revisions.
- The figure shows frequent changes in expectations, not only recession episodes. This matters because the paper’s tests use continuous changes in beliefs rather than a simple bad-times dummy.

Interpretation:

- The table is the map for the entire paper. Once the constrained side is known, the predicted return sign is mechanically different across anomaly groups.
- The classification also helps distinguish the authors’ mechanism from broad macro-risk explanations. A broad risk story would usually predict the same sign across anomaly portfolios; this paper predicts opposite signs because constrained firms appear on opposite sides of different long-short strategies.
- Figure 1 should be read as a belief shock rather than a realized productivity shock. The variable can move before realized macro conditions are visible, which is why it is useful for studying overreaction.
- Large positive revisions are periods when optimism about productivity improves; the mechanism predicts that constrained firms become especially overvalued. Large negative revisions are periods when pessimism rises; constrained firms should be less overpriced and can subsequently perform better.

Economic message: The table defines the cross-sectional sign logic, while the figure supplies the time-series belief shock. The identification comes from combining the two: the same macro-perception variable should move different anomaly spreads in opposite directions depending on which leg contains constrained firms.

Table 1
Stock market anomalies.
The table displays the 12 prominent anomalies that this paper tries to link to biased beliefs and financial constraints. The second column lists the acronyms that are used in subsequent tables to refer to the anomalies. The third column lists the literature that attributes the anomaly to biased beliefs. The fourth column lists the literature that links the anomaly characteristic to financial constraints or their major components (size, age, and profitability).

Anomaly	Acronym	Evid. of biased beliefs	Evid. of financial constraints
Panel A: Short leg more financially constrained			
Return on assets	ROA	Stambaugh et al. (2012)	Lian and Ma (2021)
Return on equity	ROE	Chen et al. (2023)	Lian and Ma (2021)
Operating profitability	OP	Chen et al. (2023)	Lian and Ma (2021) and Whited and Wu (2006)
Quality	Quality	Anness et al. (2019)	Based on profitability, profitability growth, and safety
Failure probability	FProb	Stambaugh et al. (2012)	Hoberg and Maksimovic (2015)
Ohlson's O-score	O-Score	Stambaugh et al. (2012)	Hoberg and Maksimovic (2015)
Total volatility	TV	Phlaeger et al. (2020)	Dagupta et al. (2011) and Lin and Paravisini (2013)
Idiosyncratic volatility	IVOL	Hou and Loh (2015) and Stambaugh et al. (2015)	Jiang et al. (2009)
Panel B: Long leg more financially constrained			
Book-to-market equity	BM	Lakonishok et al. (1994)	Fama and French (1995) and Maio (2014)
Long-term reversals	LTR	De Bondt and Thaler (1985)	Lamont et al. (2001) and Maio (2014)
Asset growth	IA	Cooper et al. (2008)	Cooper et al. (2008)

Table 1: stock-market anomalies and financial-constraint classification

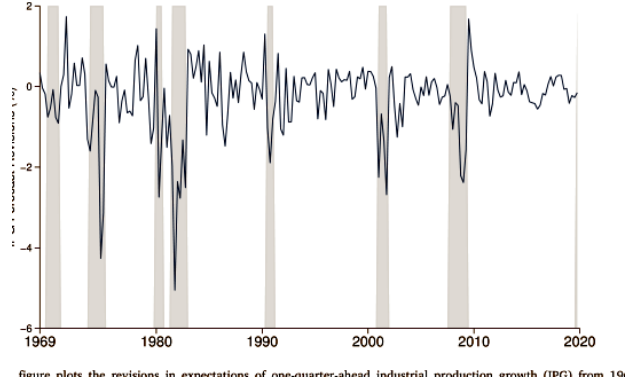


Figure 1: IPG forecast revisions

5.2 Table 2 and Table 3: anomaly summaries and key variables

Read:

- Table 2 reports average returns and financial-constraint differences for the anomaly legs. It confirms that the anomalies are economically meaningful and that the constrained-unconstrained split is not arbitrary.
- Panel A shows anomalies for which the short leg is more constrained. These are the strategies predicted to earn higher spreads after upward IPG forecast revisions.
- Panel B shows anomalies for which the long leg is more constrained. These are the strategies predicted to weaken after upward IPG forecast revisions.
- Table 3 reports summary statistics and correlations for IPG forecast revisions and competing predictors. The IPG revision variable has limited persistence and low correlation with many standard return predictors.
- The autocorrelation of IPG revisions is about 0.35, much smaller than highly persistent variables such as sentiment. This reduces the concern that IPG revisions simply proxy for a slow-moving sentiment state.

Interpretation:

- Table 2 validates the mechanism's cross-sectional premise: anomaly legs differ systematically in financial constraints. Without this fact, the sign-reversal predictions would have no empirical foundation.
- The return columns in Table 2 are useful benchmarks. They show the unconditional profitability of the anomalies before conditioning on macro beliefs. Later tables ask how those unconditional spreads change after upward or downward forecast revisions.
- The financial-constraint columns should be read directionally: the authors are not relying on a single noisy measure, but on composite rankings based on multiple constraint indices.
- Table 3 is a "separation" table. If IPG forecast revisions were highly correlated with sentiment, volatility, value spread, or factor returns, the later predictability tests would be less convincing. Low correlations make the macro-perception channel more distinct.

Economic message: Table 2 establishes which firms are likely to be most sensitive to macro belief changes. Table 3 shows that the belief variable has independent variation rather than merely repackaging standard predictors.

5.3 Table 4 and Table 5: two-regime anomaly returns

Read:

Table 2

Summary statistics for the anomalies.

The table reports summary statistics for the anomalies with more financially constrained short legs (long legs) in Panel A (Panel B). Quarterly excess returns, CAPM alphas, and Fama and French (2015) five-factor alphas are displayed in percentages. For the annual composite financial constraint index (FC index), we first build a firm-level FC index as the average score of cross-sectional ranks of the WW index (Ivovt and Wu, 2008), the IP index (Bridges and Pierce, 2010), and the IZ index (Lambert et al., 2005) for the fiscal year ending in calendar year $t-1$ and then aggregate up to the portfolio level (which is formed at the beginning of each year t) by taking the median of long-leg firms minus the median of short-leg firms. Average is the portfolio that equally combines the anomalies in each panel. The sample period spans 1990Q1 to 2019Q4 for all but the future probability anomaly (FPb), whose data begin in 1992Q1. *Novy and Vost (1987) right-leg adjusted t-statistics* are in parentheses.

Anomaly	Excess return		CAPM alpha		FF5 alpha		FC index		Long-Short	
	Long	Short	Long	Short	Long	Short	Long	Short	Long	Short
Panel A: Short leg more financially constrained										
ROA	1.92 (2.99)	1.15 (1.22)	0.77 (1.26)	0.26 (0.86)	-1.30 (-2.25)	1.56 (2.51)	0.68 (2.45)	0.22 (0.53)	-0.31 (-13.50)	0.56 (16.04)
ROE	1.89 (3.00)	1.49 (1.75)	8.29 (0.42)	0.30 (0.72)	-0.84 (-1.41)	1.94 (1.43)	0.07 (0.34)	0.51 (1.17)	-4.45 (-5.78)	0.58 (14.84)
OP	1.93 (2.95)	0.32 (0.35)	1.62 (2.47)	0.27 (0.86)	-2.00 (-3.53)	2.27 (2.45)	0.76 (2.83)	-0.80 (-1.85)	1.55 (3.12)	-0.26 (-13.57)
Quality	2.34 (3.60)	0.35 (0.38)	1.99 (2.76)	0.72 (2.49)	-2.16 (-3.96)	2.88 (4.37)	1.23 (4.80)	-1.48 (-3.54)	2.71 (5.21)	-0.34 (-13.17)
FPb	2.51 (4.17)	0.42 (0.32)	2.09 (1.91)	0.59 (2.53)	-3.36 (-3.71)	3.95 (4.12)	0.74 (1.97)	-2.15 (-3.00)	2.49 (3.12)	-0.62 (-8.89)
O-score	1.71 (2.96)	1.30 (1.42)	0.41 (0.65)	0.02 (0.06)	-1.17 (-2.34)	1.19 (1.94)	0.79 (2.56)	-0.38 (-0.93)	1.17 (2.25)	-0.42 (-7.44)
TV	1.55 (2.95)	-0.09 (-0.08)	1.64 (1.74)	0.38 (1.44)	-3.00 (-4.43)	3.39 (3.91)	-0.35 (-1.31)	-1.37 (-2.95)	1.02 (1.82)	-0.55 (-5.99)
IVOL	1.76 (3.39)	-0.26 (-0.23)	2.01 (2.17)	0.48 (1.93)	-3.06 (-4.66)	3.54 (4.18)	-0.28 (-1.40)	-1.50 (-3.13)	1.21 (2.30)	-0.62 (-5.26)
Average	1.90 (3.22)	0.54 (0.56)	1.36 (1.96)	0.36 (1.60)	-2.06 (-3.76)	2.42 (3.68)	0.45 (2.37)	-0.82 (-2.30)	1.27 (3.12)	-0.42 (-22.82)
Panel B: Long leg more financially constrained										
BM	2.72 (3.87)	1.53 (2.15)	1.19 (1.44)	0.90 (1.56)	-0.29 (-0.83)	1.18 (1.40)	0.25 (0.57)	0.27 (1.13)	-0.02 (-0.03)	0.24 (0.00)
LTR	2.43 (2.94)	1.82 (2.34)	0.62 (0.69)	0.22 (0.30)	-0.27 (-0.79)	0.49 (0.55)	-0.08 (-0.14)	0.49 (1.47)	-0.57 (-0.90)	0.44 (7.02)
IA	2.29 (3.49)	1.14 (1.56)	0.35 (2.09)	0.38 (0.89)	-1.04 (-2.76)	1.42 (2.46)	-0.21 (-0.66)	0.01 (0.05)	-0.22 (-0.57)	0.42 (31.21)
SIZE	1.99 (2.25)	1.41 (2.70)	0.38 (0.45)	0.28 (0.54)	-0.29 (-0.39)	0.11 (0.43)	0.09 (0.34)	0.22 (0.20)	-0.12 (-0.47)	0.51 (34.54)
Average	2.36 (3.39)	1.52 (2.25)	0.84 (1.25)	0.30 (0.55)	-0.37 (-1.50)	0.67 (0.97)	0.01 (0.05)	0.25 (1.56)	-0.23 (-0.85)	0.39 (19.96)

Table 3

Summary statistics for key variables.

The table displays the summary statistics. Panel A reports summary information of the key variables. Prev-IPG is the revision in consensus forecast of one-quarter-ahead industrial production growth from the Survey of Professional Forecasters, computed following Burdalo et al. (2020). Up-IPG is the indicator of an upward revision in expectations of one-quarter-ahead industrial production growth. MktRF is the value-weighted return on the CRSP portfolio of US stocks in excess of the risk-free rate. SMB, HML, RMW, and CMA are the Fama and French (2015) size, value, profitability, and investment factors, respectively. MktRF, SMB, HML, RMW, and CMA are from Kenneth French's website and compounded to quarterly percentages using monthly data. Sentiment is the quarterly average Baker and Wurgler (2004) investor sentiment index, taken from Jeffrey Wurgler's website. ρ is the first-order autocorrelation coefficient. Panel B reports the pairwise correlations between the time series, with p -values displayed in brackets. The sample is quarterly and spans from 1960Q1 to 2019Q4.

Panel A: Summary information						
Obs	Mean	Std. Dev.	Median	Min	Max	ρ
Prev-IPG	204	0.90	-0.12	-5.05	1.74	0.35
Up-IPG	204	0.41	0.49	0.00	1.00	0.28
MktRF	204	1.65	8.60	2.69	-26.46	0.05
SMB	204	0.38	5.38	0.12	-12.38	-0.05
HML	204	0.96	5.99	0.57	-16.83	0.13
RMW	204	0.89	4.29	0.62	-14.60	0.12
CMA	204	0.93	4.07	0.24	-7.95	0.08
Sentiment	204	0.02	0.97	-0.01	-2.38	0.95
Panel B: Pairwise correlations						
	Prev-IPG	MktRF	SMB	HML	RMW	CMA
Prev-IPG	1.00					
MktRF	0.03 [0.69]	1.00				
SMB	0.03 [0.30]	0.41 [0.00]	1.00			
HML	0.06 [0.38]	-0.32 [0.00]	0.01 [0.91]	1.00		
RMW	0.03 [0.87]	-0.28 [0.00]	-0.19 [0.01]	0.10 [0.16]	1.00	
CMA	0.03 [0.72]	-0.42 [0.00]	-0.09 [0.00]	0.75 [0.27]	0.08 [0.37]	1.00
Sentiment	-0.02 [0.73]	-0.10 [0.15]	-0.13 [0.04]	0.11 [0.13]	0.26 [0.00]	0.09 [0.18]

- Table 4 splits quarters into upward and downward IPG forecast revisions and reports raw excess returns.
- For anomalies with constrained short legs, the average long-short return is 3.73% after upward IPG revisions and -0.35% after downward revisions.
- For anomalies with constrained long legs, the average long-short return is -1.30% after upward revisions and 2.34% after downward revisions.
- The long-leg and short-leg columns help reveal where the spread comes from. The mechanism predicts that the constrained side should be the side most negatively affected after upward revisions.
- Table 5 repeats the same two-regime test using CAPM-adjusted returns rather than raw excess returns.
- The sign pattern remains after market adjustment, so the result is not just market beta variation across regimes.

Interpretation:

- These are the main return tables. The key object is not only the average return in each regime, but the difference between the upward- and downward-revision regimes.
- The two panels should be read as mirror images. In Panel A-type anomalies, constrained firms are in the short leg, so constrained underperformance helps the long-short strategy. In Panel B-type anomalies, constrained firms are in the long leg, so constrained underperformance hurts the long-short strategy.
- The negative or weak returns in the “wrong” regime are important. They show that anomalies are not constant premiums; their payoffs vary with macro perceptions.
- CAPM adjustment in Table 5 matters because it addresses a simple alternative interpretation: perhaps upward revisions just increase or reduce market exposure. The persistence of the pattern after CAPM adjustment points to anomaly-specific predictability.
- Economically, the tables imply that factor timing based on macro expectations is not symmetric across all anomalies. The right timing rule depends on the constraint location within each anomaly portfolio.

Economic message: The two-regime evidence is the core return result. Macro optimism appears to lower subsequent returns of financially constrained firms; whether that strengthens or weakens an anomaly depends on whether constrained firms are shorted or bought.

5.4 Figure 2 and Table 6: alternative surveys and predictive regressions

Read:

Table 4
Annually returns and productivity forecast revisions.
The table reports the average quarterly excess returns for anomalies following upward and downward revisions in expectations of industrial production growth (IPG). The average returns following up and downward forecast revisions are estimates of α_U and α_D in the regression $R_{it} = \alpha_U \cdot 1_{U_t} + \alpha_D \cdot 1_{D_t} + \beta_{IPG} \cdot \Delta IPG_{it} + \epsilon_{it}$, where 1_U and 1_D are, respectively, indicators for quarters with upward and downward revisions in IPG expectations from the SPF survey, and R_{it} is the excess return on the long leg, the short leg, or the difference. Panel A reports results for anomalies with more financially constrained short legs, and Panel B reports results for anomalies with more financially constrained long legs. *Average* is the return to the portfolio that equally combines the anomalies in each panel. The sample period spans from 1969Q1 to 2019Q4 for all but the failure probability anomaly (FProb), whose data begin in 1976Q1. Newey-West eight-lag adjusted *t*-statistics are reported in parentheses.

Anomaly	Long leg		Short leg		Up-Down		Long-short		
	Up	Down	Up-Down	Up	Down	Up-Down	Up	Down	
Panel A: Short leg more financially constrained									
ROA	1.11 (1.14)	2.53 (2.95)	-1.42 (-1.13)	-1.73 (-1.23)	3.25 (2.53)	-4.97 (-2.70)	2.84 (3.74)	-0.72 (-0.88)	3.55 (3.63)
ROE	0.99 (1.09)	2.59 (2.95)	-1.61 (-1.38)	-1.49 (-1.66)	3.85 (3.00)	-5.34 (-2.89)	2.48 (2.86)	-1.26 (-1.40)	3.74 (3.50)
OP	1.27 (1.25)	2.45 (2.78)	-1.18 (-0.90)	-2.71 (-2.11)	2.53 (1.97)	-5.24 (-2.98)	3.97 (5.47)	-0.08 (-0.10)	4.06 (4.49)
Quality	1.71 (1.74)	2.82 (3.37)	-1.11 (-0.90)	-2.47 (-1.78)	2.43 (1.79)	-4.50 (-2.64)	4.18 (4.99)	0.39 (0.40)	3.79 (3.27)
FProb	1.82 (1.90)	3.00 (3.49)	-1.18 (-0.89)	-3.65 (-1.97)	3.30 (1.97)	-6.96 (-2.98)	5.48 (4.08)	-0.30 (-0.21)	5.78 (3.37)
O-Score	0.86 (0.81)	2.34 (2.68)	-1.49 (-1.10)	-1.88 (-1.49)	3.65 (2.75)	-5.53 (-3.04)	2.73 (3.36)	-1.31 (-1.44)	4.04 (3.76)
TV	1.29 (1.21)	1.78 (2.45)	-0.49 (-0.53)	-2.71 (-1.84)	1.87 (1.23)	-4.58 (-2.18)	4.00 (3.49)	-0.09 (-0.07)	4.09 (2.53)
IVOL	1.43 (2.21)	2.02 (2.83)	-0.59 (-0.59)	-2.65 (-1.87)	1.52 (1.03)	-4.18 (-2.09)	4.08 (3.50)	0.50 (0.41)	3.58 (2.34)
Average	1.36 (1.48)	2.39 (2.94)	-1.13 (-0.98)	-2.46 (-1.84)	2.74 (2.05)	-5.20 (-2.84)	3.73 (4.90)	-0.35 (-0.38)	4.07 (3.97)
Panel B: Long leg more financially constrained									
BM	0.51 (0.58)	4.34 (3.60)	-3.83 (-2.37)	-3.83 (-2.37)	1.79 (0.75)	-2.32 (-2.02)	2.24 (2.11)	-2.52 (-2.02)	2.26 (2.08)
LTR	-0.58 (-0.54)	4.61 (3.41)	-3.19 (-2.82)	-3.19 (-2.82)	2.46 (0.87)	-1.41 (-0.86)	2.15 (1.85)	-3.79 (-3.31)	2.00 (1.76)
IA	-0.34 (-0.38)	4.20 (3.98)	-4.54 (-3.16)	-4.54 (-3.16)	0.20 (0.17)	-1.72 (-0.70)	2.29 (2.25)	-2.82 (-3.19)	2.73 (2.70)
SIZE	-1.70 (-1.31)	4.70 (3.74)	-6.40 (-3.72)	-6.40 (-3.72)	1.05 (1.27)	-2.03 (-0.88)	2.67 (2.49)	-5.43 (-5.11)	1.60 (1.42)
Average	-0.58 (-0.58)	4.46 (3.89)	-4.99 (-3.19)	-4.99 (-3.19)	0.77 (0.75)	-2.13 (-2.28)	2.34 (2.78)	-3.64 (-4.73)	2.15 (2.50)

Table 5
CAPM-adjusted anomaly returns and productivity forecast revisions.
The table reports the average quarterly CAPM-adjusted returns for anomalies following upward and downward revisions in expectations of industrial production growth (IPG). The average returns following up and downward forecast revisions are estimates of α_U and α_D in the regression $R_{it} = \alpha_U \cdot 1_{U_t} + \alpha_D \cdot 1_{D_t} + \beta \cdot Mkt_{F,t} + \epsilon_{it}$, where 1_U and 1_D are, respectively, indicators for quarters with upward and downward revisions in IPG expectations from the SPF survey, $Mkt_{F,t}$ is the market factor, and R_{it} is the excess return on the long leg, the short leg, or the difference. Panel A reports results for anomalies with more financially constrained short legs, and Panel B reports results for anomalies with more financially constrained long legs. *Average* is the return to the portfolio that equally combines the anomalies in each panel. The sample period spans from 1969Q1 to 2019Q4 for all but the failure probability anomaly (FProb), whose data begin in 1976Q1. Newey-West eight-lag adjusted *t*-statistics are reported in parentheses.

Anomaly	Long leg		Short leg		Up-Down		Long-short	
	Up	Down	Up-Down	Up	Down	Up-Down	Up	Down
ROA	0.44 (1.10)	0.11 (0.31)	-0.33 (-0.37)	-0.27 (-0.37)	-2.43 (-2.87)	3.14 (4.62)	0.38 (0.48)	2.75 (3.10)
ROE	0.31 (0.94)	0.14 (0.39)	-0.17 (-0.42)	-2.46 (-3.41)	0.35 (0.44)	-2.80 (-3.08)	2.77 (3.47)	-0.19 (-0.21)
OP	0.60 (1.37)	0.02 (0.06)	-0.57 (-1.20)	-3.63 (-5.91)	-0.79 (-1.16)	-2.83 (-4.06)	4.22 (6.03)	0.82 (1.01)
Quality	1.06 (2.56)	0.46 (1.34)	-0.59 (-1.23)	-3.47 (-5.40)	-1.19 (-1.63)	-2.28 (-2.64)	4.53 (5.91)	1.65 (1.88)
FProb	0.87 (1.95)	0.37 (1.42)	-0.50 (-0.93)	-4.49 (-5.79)	-1.77 (-2.91)	-3.72 (-5.68)	5.36 (5.68)	2.15 (1.73)
O-Score	0.18 (0.36)	-0.13 (-0.37)	0.30 (0.91)	-2.86 (-4.75)	0.09 (0.14)	-2.95 (-3.47)	3.03 (1.67)	-0.22 (-0.36)
TV	0.82 (2.44)	0.07 (0.23)	0.74 (2.03)	-3.88 (-4.89)	-2.35 (-2.38)	-1.53 (-1.23)	4.70 (5.08)	2.42 (2.61)
IVOL	0.92 (2.58)	0.16 (0.37)	0.75 (1.89)	-3.77 (-4.37)	-2.53 (-2.69)	-1.24 (-1.00)	4.69 (4.74)	2.00 (2.32)
Average	1.26 (2.20)	0.44 (0.56)	-0.49 (-1.49)	-3.50 (-6.20)	-1.00 (-1.40)	-2.49 (-3.20)	4.14 (6.43)	1.15 (3.44)
Panel B: Long leg more financially constrained								
BM	-0.21 (-0.35)	1.71 (2.08)	-1.92 (-2.07)	0.56 (0.13)	-0.55 (-1.37)	0.61 (1.17)	-0.28 (-0.28)	-2.53 (-2.02)
LTR	-0.58 (-2.34)	4.61 (1.43)	-3.19 (-3.01)	-3.19 (-3.01)	-2.82 (-0.46)	-2.00 (1.21)	-3.68 (-2.04)	-3.68 (1.76)
IA	-0.34 (-0.38)	4.20 (3.98)	-4.54 (-3.16)	-4.54 (-3.16)	0.20 (0.17)	-1.72 (-0.70)	2.29 (2.25)	-2.82 (-3.19)
SIZE	-1.70 (-1.31)	4.70 (3.74)	-6.40 (-3.72)	-6.40 (-3.72)	1.05 (1.27)	-2.03 (-0.88)	2.67 (2.49)	-5.43 (-5.11)
Average	-0.53 (-0.58)	4.46 (3.89)	-4.99 (-3.19)	-4.99 (-3.19)	0.77 (0.75)	-2.13 (-2.28)	2.34 (2.78)	-3.64 (-4.73)

- Figure 2 compares SPF macro expectation revisions with Michigan consumer survey revisions and Duke CFO survey optimism.
- The series move in broadly similar directions, suggesting that the SPF variable captures a wider change in macro perceptions rather than a narrow professional-forecaster artifact.
- Table 6 estimates predictive regressions using standardized IPG forecast revisions rather than a simple up/down split.
- For the average portfolio with constrained short legs, the long-short coefficient is positive: 2.04 for excess returns and 1.37 for CAPM alphas.
- For the average portfolio with constrained long legs, the long-short coefficient is negative: -2.28 for excess returns and -2.21 for CAPM alphas.
- Because the predictor is standardized, the coefficients can be read as the return effect of a one-standard-deviation revision in expected IPG.

Interpretation:

- Figure 2 strengthens the behavioral interpretation. If CFOs, consumers, and professional forecasters move together, the SPF measure is more plausibly capturing a market-wide perception state.
- Table 6 is a stronger version of the regime test because it uses the magnitude of the revision. A large positive revision should have a larger effect than a small positive revision.
- The market-factor control means the coefficient is not simply saying that macro optimism predicts the market. It is saying that macro optimism predicts the relative payoff of anomaly legs.
- The opposite signs in the average portfolio rows are especially important because they compress many anomaly-specific tests into the paper's central mechanism.
- The regression format also makes the results easier to compare with other return-predictability variables and later horse-race tests.

Economic message: Figure 2 supports the interpretation of IPG revisions as broad macro beliefs. Table 6 shows that the size of the belief revision matters and that the sign of predictability follows the financial-constraint location.

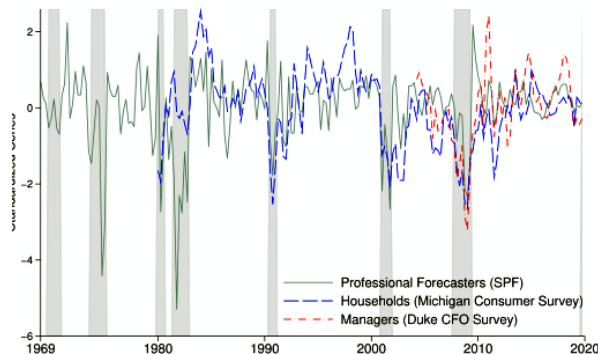


Figure 2: SPF, consumer survey, and CFO survey revisions

Table 6
Predictive regressions.
The table reports estimates of β in the following predictive regression:
$$R_{i,t+1} = \alpha + \beta \text{ForecastIPG}_i + \gamma \text{MarketFactor}_{i,t+1} + \epsilon_{i,t+1}$$
where ForecastIPG_i is the revision in expectations of industrial production growth from the SPF survey, $\text{MarketFactor}_{i,t+1}$ is the quarterly excess return on the long leg, the short leg, or the difference. Panel A reports results for anomalies with more financially constrained short legs, and Panel B reports results for anomalies with more financially constrained long legs. *Average* is the return to the portfolio that equally combines the anomalies in each panel. Columns (1) to (3) report results for excess returns, and columns (4) to (6) report results for CAPM-adjusted returns. All regressions include standardized independent variables with zero mean and unit variance. The sample period spans from 1969Q1 to 2019Q4 for all but the failure probability anomaly (FProb), whose data begin in 1976Q1. Newey-West eight-lag adjusted t -statistics are reported in parentheses.

Anomaly	Excess return			CAPM alpha		
	Long leg (1)	Short leg (2)	Long-Short (3)	Long leg (4)	Short leg (5)	Long-Short (6)
Panel A: Short leg more financially constrained						
ROA	-1.07 (-1.44)	-2.71 (-2.53)	1.64 (2.06)	0.02 (0.09)	-1.12 (-2.41)	1.14 (2.44)
ROE	-1.11 (-1.25)	-3.09 (-2.53)	1.98 (3.27)	-0.00 (-0.01)	-1.51 (-3.16)	1.50 (2.77)
OP	-1.02 (-1.33)	-2.61 (-2.37)	1.60 (3.00)	0.07 (0.33)	-1.12 (-3.31)	1.19 (2.76)
Quality	-1.03 (-1.52)	-2.87 (-1.95)	1.83 (1.88)	0.03 (0.10)	-1.24 (-2.29)	1.27 (1.75)
FProb	-0.34 (-0.60)	-2.67 (-1.27)	2.32 (1.24)	0.14 (0.39)	-1.73 (-1.39)	1.86 (1.24)
O-Score	-1.07 (-1.45)	-2.90 (-1.95)	1.82 (2.07)	0.04 (0.18)	-1.29 (-2.66)	1.33 (2.12)
TV	-0.48 (-0.88)	-2.97 (-1.49)	2.49 (1.46)	0.29 (1.22)	-1.07 (-1.88)	1.37 (1.13)
IVOL	-0.50 (-0.92)	-2.83 (-1.49)	2.33 (1.48)	0.34 (1.49)	-1.01 (-1.08)	1.35 (1.19)
Average	-0.89 (-1.31)	-2.92 (-1.98)	2.04 (2.11)	0.13 (0.65)	-1.24 (-2.19)	1.37 (1.99)
Panel B: Long leg more financially constrained						
BM	-2.33 (-1.96)	-1.15 (-0.90)	-1.45 (-2.42)	0.32 (-2.26)	-1.47 (0.99)	-1.47 (-2.30)
LTR	-4.24 (-2.99)	-0.85 (-0.82)	-3.39 (-3.20)	-2.83 (-3.37)	0.53 (1.37)	-3.35 (-3.34)
IA	-2.74 (-2.42)	-1.51 (-1.19)	-1.23 (-3.13)	-1.51 (-4.02)	-0.08 (-0.19)	-1.43 (-3.73)
SIZE	-3.77 (-2.63)	-0.70 (-0.98)	-3.06 (-3.65)	-2.10 (-4.08)	0.26 (2.85)	-2.59 (-4.14)
Average	-3.27 (-2.41)	-0.99 (-1.00)	-2.28 (-4.25)	-1.95 (-3.87)	0.26 (1.04)	-2.21 (-4.33)

Table 6: predictive regressions

5.5 Table 7: controlling for confounding variables

Read:

- Table 7 controls for sentiment, surplus consumption ratio, degree of extrapolative weighting, return dispersion, value spread, and realized volatility.
- The IPG forecast-revision coefficient generally keeps the predicted sign after these controls.
- The control variables do not reproduce the same cross-anomaly sign pattern as IPG forecast revisions.
- This table asks whether the main predictor is merely a disguised version of standard behavioral, risk, or market-state variables.

Interpretation:

- This is the main horse-race table. The most important reading is whether the IPG coefficient survives after adding one competing explanation at a time.
- Sentiment controls address the concern that the results are simply investor mood. The survival of the IPG effect suggests that macro productivity expectations are more targeted than general sentiment.
- Surplus consumption and volatility controls address risk-based explanations. If the pattern were only compensation for changing risk aversion or aggregate uncertainty, those controls should substantially absorb the coefficient.
- Value spread and return dispersion controls address time-varying anomaly profitability. The continued IPG effect suggests that the mechanism is not just standard factor-cycle timing.
- The table is most persuasive because the same predictor keeps the correct sign in two anomaly groups with opposite predicted directions.

Economic message: The forecast-revision effect is not simply sentiment, risk aversion, value spread, return dispersion, or volatility in disguise. It behaves like a distinct macro-belief channel interacting with financial constraints.

Table 7
Predictive regressions: Controlling for confounding variables.
The table reports estimates of β and γ in the following bivariate predictive regression:

$$R_{i,t+1} = \alpha + \beta \text{FrevIPG}_i + \gamma z_i + \epsilon_{i,t+1}$$
where FrevIPG_i is the revision in expectations of industrial production growth, z_i is the control variable, and $R_{i,t+1}$ is the quarterly excess return on the long-short anomalies. The control variable z_i is the average Baker and Wurgler (2006) investor sentiment index in columns (1) and (2), the surplus consumption ratio calculated following Wachter (2006) in columns (3) and (4), the Casella and Gulen (2018) degree of extrapolative weighting (DOX) in columns (5) and (6), the Stivers and Sun (2010) return dispersion in columns (7) and (8), the anomaly's value spread $\log(\text{BM}_{\text{long}}) - \log(\text{BM}_{\text{short}})$ in columns (9) and (10), and the realized volatility of daily anomaly returns in columns (11) and (12). All regressions include standardized independent variables with zero mean and unit variance. Panels A and C (Panels B and D) report results for anomalies with more financially constrained short legs (long legs). Average is the portfolio that equally combines the anomalies in each panel. The sample period spans from 1969Q1 to 2019Q4, and the DOX data, obtained from Cooper et al. (2024), ends in 2018Q4. Newey-West eight-lag adjusted t -statistics are reported in parentheses.

$z_i =$	Sentiment		Surplus ratio		DOX	
	β (1)	γ (2)	β (3)	γ (4)	β (5)	γ (6)
Panel A: Short leg more financially constrained						
ROA	1.69 (3.32)	2.25 (4.29)	1.66 (3.02)	0.51 (0.89)	1.70 (3.26)	0.21 (0.25)
ROE	2.03 (3.49)	2.27 (3.86)	1.99 (3.22)	0.36 (0.65)	2.06 (3.38)	0.30 (0.31)
OP	1.65 (3.50)	1.98 (5.11)	1.60 (2.95)	0.04 (0.07)	1.27 (2.93)	-0.11 (-0.17)
Quality	1.89 (2.31)	2.41 (4.70)	1.85 (1.86)	0.28 (0.39)	1.92 (2.08)	0.28 (0.33)
FProb	2.92 (1.68)	5.52 (4.48)	2.35 (1.23)	0.55 (0.41)	2.72 (1.48)	1.13 (0.84)
O-Score	1.87 (2.62)	2.01 (3.88)	1.85 (2.03)	0.71 (1.20)	1.79 (2.14)	-0.10 (-0.16)
TV	2.58 (1.71)	3.40 (4.17)	2.54 (1.45)	0.96 (1.24)	2.77 (1.75)	0.98 (0.82)
IVOL	2.41 (1.75)	3.17 (3.93)	2.38 (1.47)	1.02 (1.38)	2.52 (1.69)	0.67 (0.56)
Average	2.11 (2.56)	2.65 (4.86)	2.07 (2.07)	0.56 (0.90)	2.16 (2.38)	0.40 (0.45)
Panel B: Long leg more financially constrained						
BM	-1.46 (-2.46)	-0.46 (-0.63)	-1.41 (-2.42)	0.97 (1.30)	-1.07 (-1.53)	1.26 (1.80)
LTR	-3.41 (-3.35)	-0.68 (-0.78)	-3.36 (-3.25)	0.65 (0.88)	-3.04 (-2.50)	1.27 (1.67)
IA	-1.23 (-3.14)	0.17 (0.29)	-1.22 (-3.04)	0.36 (0.81)	-1.10 (-2.35)	0.49 (0.72)
SIZE	-3.11 (-4.25)	-1.80 (-2.29)	-3.06 (-3.62)	0.09 (0.13)	-2.81 (-3.28)	0.90 (1.20)
Average	-2.30 (-4.63)	-0.69 (-1.13)	-2.26 (-4.33)	0.52 (0.94)	-2.00 (-3.17)	1.00 (1.58)
Panel C: Short leg more financially constrained						
ROA	1.74 (2.90)	0.48 (0.66)	1.50 (2.64)	0.81 (1.23)	1.69 (2.93)	0.65 (0.81)
ROE	2.02 (3.06)	0.19 (0.21)	1.73 (2.77)	1.16 (1.70)	2.02 (3.16)	0.49 (0.47)
OP	1.64 (2.84)	0.16 (0.25)	1.57 (2.80)	0.33 (0.61)	1.61 (2.90)	0.06 (0.09)
Quality	1.81 (1.83)	-0.14 (-0.14)	1.63 (1.64)	0.93 (1.38)	1.75 (1.90)	-0.75 (-0.97)
FProb	2.25 (1.23)	-0.45 (-0.23)	1.84 (1.09)	2.84 (2.20)	2.36 (1.20)	-0.20 (-0.09)
O-Score	1.77 (1.97)	-0.27 (-0.49)	1.74 (1.95)	0.49 (0.75)	1.77 (2.06)	-0.85 (-1.21)
TV	2.29 (1.40)	-0.51 (-0.31)	1.97 (1.39)	3.08 (3.33)	2.31 (1.35)	-0.96 (-0.50)
IVOL	2.29 (1.43)	-0.19 (-0.12)	1.81 (1.40)	3.33 (3.64)	2.30 (1.42)	-0.17 (-0.09)
Average	2.02 (2.03)	-0.10 (-0.09)	1.60 (1.80)	1.83 (2.45)	2.01 (2.02)	-0.19 (-0.16)
Panel D: Long leg more financially constrained						
BM	-1.19 (-2.02)	1.24 (2.27)	-1.44 (-2.36)	0.11 (0.10)	-1.39 (-2.32)	0.39 (0.58)
LTR	-3.08 (-2.97)	-1.49 (2.84)	-3.34 (-3.22)	0.70 (0.67)	-3.31 (-3.00)	0.36 (0.49)
IA	-1.08 (-2.45)	0.72 (1.57)	-1.19 (-2.95)	0.31 (0.65)	-1.17 (-2.66)	0.39 (0.63)
SIZE	-2.72 (-3.74)	1.66 (2.75)	-2.92 (-4.45)	2.03 (3.00)	-2.86 (-3.57)	1.33 (2.11)
Average	-2.02 ---	1.28 ---	-2.18 ---	0.91 ---	-2.16 ---	0.64 ---

Table 7: predictive regressions controlling for confounding variables

5.6 Figure 3 and Table 8: rotation strategy and misperception evidence

Read:

- Figure 3 compares a factor-rotation strategy with the underlying static anomaly groups.
- The strategy uses the previous quarter's IPG forecast revision to choose the anomaly group predicted to perform better.
- The cumulative return of the rotation strategy exceeds the static anomaly groups, indicating that the predictability has economically meaningful portfolio implications.
- Table 8 studies analyst long-term earnings growth forecast errors.
- Upward IPG forecast revisions are followed by forecast errors consistent with overoptimism, especially for financially constrained firms.
- For the average portfolio in Panel A, the long-short LTG forecast-error coefficient is 6.17; in Panel B, it is -3.78, again matching the location of constrained firms in the anomaly legs.

Interpretation:

- Figure 3 translates the paper's return predictability into a trading rule. It is not only a statistical result; it can be organized into a simple rotation across anomaly groups.
- The rotation strategy should be interpreted as evidence of economic magnitude, not necessarily as a fully implementable strategy after transaction costs, capacity limits, and shorting constraints.

- Table 8 is central for the behavioral mechanism. If macro optimism merely reflected better fundamentals, analyst forecasts should not be systematically too optimistic afterward. Forecast errors suggest expectations overshoot.
- The constrained-leg pattern in Table 8 connects beliefs to the cross-section. The firms most exposed to financing frictions are also the firms for which macro optimism seems to generate larger earnings-growth misperceptions.
- Together, the figure and table connect three steps: macro belief revision, forecast overreaction, and predictable anomaly returns.

Economic message: The rotation strategy shows that the return pattern has investment relevance. The LTG forecast-error evidence explains why the pattern exists: macro optimism appears to translate into overoptimistic beliefs about constrained firms' fundamentals.

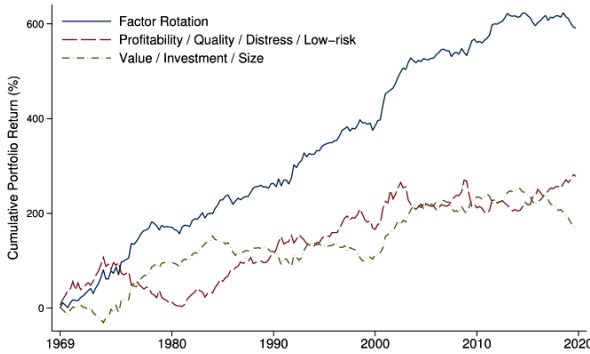


Figure 3: rotation strategy performance

Table 8

Evidence of misperception.

The table reports the estimates of β in the following regression:

$$LTG_Error_{t+1} = \alpha + \beta Frev_IPG_t + \epsilon_{t+1},$$

where LTG_Error_{t+1} is errors in analysts' forecasts of long-term earnings growth, and $Frev_IPG_t$ is the revision in expectations of one-quarter-ahead industrial production growth. LTG_Error_{t+1} is calculated as the difference between realized earnings growth and the LTG forecast, that is, $(EPS_{t+1}/EPS_{t+1})^{1/3} - LTG_{t+1}$. We restrict the calculation of realized LTG to the firms with positive EPS in quarter $t+1$, and aggregate it at the portfolio level using capitalization-weighted sum. $Frev_IPG_t$ is standardized to have zero mean and unit variance. Results are reported for anomalies with financially constrained short legs in Panel A and financially constrained long legs in Panel B. *Average* is the portfolio that equally combines the anomalies in each panel. LTG forecasts begin from 1982Q4 because of data availability. Newey-West eight-lag adjusted t -statistics are reported in parentheses.

	Long leg (1)	Short leg (2)	Long-Short (3)
Panel A: Short leg more financially constrained			
ROA	-3.23 (-3.16)	-6.66 (-3.54)	3.43 (1.57)
ROE	-1.87 (-2.62)	-10.46 (-2.90)	8.60 (2.24)
OP	-2.70 (-2.94)	-3.68 (-1.46)	0.98 (0.37)
Quality	-2.60 (-2.79)	-9.71 (-4.09)	7.11 (2.75)
FProb	-3.18 (-3.27)	-18.65 (-1.48)	15.48 (1.19)
O-Score	-3.97 (-3.06)	-6.09 (-3.25)	2.12 (0.89)
TV	-2.19 (-2.81)	-6.50 (-2.37)	4.31 (1.52)
IVOL	-1.91 (-2.46)	-7.22 (-3.35)	5.31 (2.76)
Average	-2.74 (-3.46)	-8.90 (-3.62)	6.17 (2.44)
Panel B: Long leg more financially constrained			
BM	-7.91 (-3.20)	-1.91 (-2.43)	-6.00 (-2.32)
LTR	-7.90 (-2.79)	-3.72 (-2.62)	-4.18 (-1.33)
IA	-6.73 (-2.83)	-4.09 (-3.15)	-2.64 (-1.38)
SIZE	-2.94 (-2.95)	-2.80 (-3.75)	-0.14 (-0.14)
Average	-7.43 (-2.80)	-3.65 (-3.01)	-3.78 (-1.78)

Table 8: evidence of misperception

5.7 Table 9 and Table 10: corporate activities and constraint-sorted portfolios

Read:

- Table 9 studies contemporaneous changes in bond yields, debt, long-term debt, equity, investment, and financial constraints.
- Following upward IPG revisions, financially constrained firms show larger changes in financing and investment activity.

Interpretation:

- The weak stock-market prediction is useful because it shows the paper is not mainly about aggregate equity timing. The strongest evidence is cross-sectional and credit-related.
- The negative corporate-bond return predictability is consistent with overoptimism or credit easing being followed by reversal in credit markets.
- The credit-condition results give a dynamic interpretation: optimism relaxes credit today, but this easing can be followed by later tightening.
- The unemployment-revision comparison helps show that the sign of the macro belief matters. Positive unemployment revisions are bad news, while positive IPG revisions are good-news optimism.
- Table 11 therefore broadens the mechanism from anomaly returns to credit cycles, but it also clarifies that the paper’s sharpest tests are still in the cross-section of constrained versus unconstrained firms.

Economic message: The aggregate evidence supports the overreaction and credit-cycle interpretation: optimism relaxes credit today but is followed by weaker bond returns and later tightening.

Table 11
Aggregate return predictability.
The table reports the estimation results of regressing excess market returns and bank credit conditions on revisions in expectations of aggregate productivity. $MktRF$ is the quarterly excess stock market return, calculated as the compounded monthly excess stock market return from Kenneth French’s website. $MktB$ is the quarterly excess bond market return, calculated as the compounded monthly excess bond market return from the compustat website in *Dickerson et al. (2023)*. $TightCredit$ is the percentage of bank loan officers reporting that they are tightening commercial and industrial lending standards from the Federal Reserve’s Senior Loan Officer Opinion Survey (SLOOS). $FreeIPG$ ($FreeER$) is the revision in consensus forecast of one-quarter-ahead industrial production growth (unemployment rate) from the SPF. Independent variables are standardized to have zero mean and unit variance. The sample spans from 1969Q1 to 2019Q4 for stock market returns, from 2002Q3 to 2019Q4 for corporate bond market returns, and from 1990Q2 to 2019Q4 for the SLOOS survey. Newey-West eight-lag adjusted t -statistics are reported in parentheses.

Dependent variable	$MktRF_{it}$		$MktB_{it}$		$TightCredit_{it}$		$TightCredit_{it}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$FreeIPG_t$	-1.08 (-1.35)		-1.35 (-3.21)		-14.60 (-4.71)		5.00 (1.85)
$FreeER_t$		1.46 (2.38)		1.21 (3.03)		15.94 (5.61)	-7.31 (-2.92)
Intercept	1.68 (2.80)	1.68 (2.70)	1.50 (3.75)	1.39 (4.03)	6.00 (1.44)	5.38 (1.52)	4.17 (0.90)
Obs	203	203	70	70	119	119	119

Table 11: aggregate return predictability

6 Short summary

- The paper studies how revisions in subjective macroeconomic expectations predict cross-sectional anomaly returns.
- The main perception variable is the SPF revision in expected one-quarter-ahead industrial production growth.
- The mechanism is heterogeneous: financially constrained firms are more sensitive to macro optimism and pessimism.
- For profitability, quality, distress, and low-risk anomalies, the constrained firms are in the short leg.
- For value, investment, and size anomalies, the constrained firms are in the long leg.
- Upward IPG forecast revisions strengthen the first group and weaken or reverse the second group.
- Downward IPG forecast revisions produce the opposite pattern.
- Predictive regressions show the same sign pattern using continuous standardized forecast revisions.
- The findings remain after controlling for sentiment, risk-aversion proxies, extrapolative-weighting proxies, return dispersion, value spread, realized volatility, and macro shocks.
- IPG revisions predict analyst long-term earnings growth forecast errors, consistent with overreaction.
- Constrained firms change financing and investment activity more after upward IPG revisions.
- A factor rotation strategy based on IPG forecast revisions outperforms the underlying static anomaly groups.
- Constraint-sorted portfolios show the same return timing as anomaly portfolios.
- Aggregate tests show that IPG revisions predict corporate bond market returns and credit conditions.
- The central lesson is that macro perceptions interact with financial constraints to create predictable time variation in anomalies.

7 Conclusion

7.1 Core conclusion

He, Su, and Yu show that subjective macroeconomic productivity expectations matter for anomaly returns because their effects are stronger for financially constrained firms. An upward revision in expected productivity can make constrained firms' prices rise too much, leading to lower subsequent returns. The sign of the anomaly response depends on whether constrained firms are in the long leg or short leg.

7.2 Economic intuition

The intuition is that constrained firms are more sensitive to changes in perceived financing conditions. Good macro news relaxes perceived constraints and encourages financing and investment, but it can also generate overoptimistic prices. When expectations later disappoint or mean-revert, constrained firms experience stronger reversals. This produces predictable variation in standard anomalies.

7.3 Takeaway

The paper connects three ideas: subjective macroeconomic beliefs, financial constraints, and anomaly returns. Its main message is not simply that macro news predicts returns, but that macro *perceptions* predict different anomalies with opposite signs depending on which portfolio leg contains financially constrained firms.